MICROSOFT SQL SERVER (SQL + T-SQL)

Overview:

Microsoft SQL Server is one the most popular Relational Database Management System (RDBMS) used in Microsoft universe. It can be used for data storage as well as for data reading for applications which can be either on desktop or API/Browse.

Course Objectives: Learn about SQL–Structured Query Language

- Build database using Data Definition Language Statements Perform basic CRUD operations using Data Manipulation Language statements like Insert, Update and Delete Write and call Stored Procedures, Functions, Triggers, Cursors and Optimizations techniques in a database.

Pre-requisite / Target Audience:

This sql server course can be taken by any beginner who wants to build career in Information Technology. The subscriber needs to have working knowledge of Windows Operating System.

Module 1:- Introduction to Basic Database Concepts

- Overview of Data, Field, Record, Files, File Groups and database
- Limitations of File Management System.
- Exploring Relational DBMS VS RDBMS
- Understanding Client And Server Architecture

Module 2:- Normalization

- First Normal Form, Second Normal Form, Third Normal and EF.Cod rules Form Practically Normalizing Tables

Module 3:- Introduction to SQL Server

- SQL Server Version history and different editions
- Overview of Components, Tools and Software Installations
- Starting and Stopping SQL Server Instances / Services
- Introduction to Management Studio
- Types of System Databases in SQL (Master, Model, MSDB, TempDB)

Module 4: Introduction to SQL

- SQL Commands (DDL,DML,DQL,TCL,DCL)
- Data types, Operators types of Operators
- Exploring Commands with Examples

Module 5: Working with Constraints & Data Integrity

- Primary Key, Unique, NULL, NOT NULL,CHECK, Default, Composite Key, Candidate Key and Foreign key
- Real time scenarios to create database and Objects

Module 6: Working with Queries (DQL)

- Understanding Select Statement \ Query Processing steps
- Clauses (TOP, WHERE, GROUP BY, HAVING, Order By with Offset and fetch) with Operators
- CASE and IIF Statement

Module 7: Aggregate Functions / String Functions / Date & Time Functions

- Using String functions in Queries
- Count, Sum, Min, Max, Avg, Group by and Having Clauses
- Date time Functions
- Windows Functions / Analytical Functions
- with Grouping sets, Rollup and Cube
- RANK Functions

Module 8: Set Operators

- Introduction of Set Operators and Types
- UNION, UNION ALL, INTERSECT, EXCEPT

Module 9: Sub Query's

- Introduction of Sub query's
- Sub query's, Co-Related Sub Query's

Module 10: Joins

- Introduction to Joins and types
- Inner Join / Left Outer Join / Right Outer Join / Full Outer Join /Self-Join /Cross Join

Module 11: Table Expressions / Temporary tables

- Derived Tables , Table variables, CTE (Common Table Expressions),Temporary Tables (Local,Global)
- PIVOT / UNPIVOT, Temporal tables

Module 12: Working with Indexes

- Heap structure vs. B-Tree Structure Format
- Types Of indexes
- Clustered Index, Non-Clustered Index, Filter Index, Covering Index, Column Store index, Index Scan, Index Seek, Index Re-Organize and Index -Rebuild

Module 13: Implementing Views & partitions

- Introduction & Advantages of Views
- Types of Views (Simple / Complex / Materialized)
- Creating, Altering, Dropping Views
- SQL Server Catalogue Views (System Defined Views)

Module 14: Transactions

- ACID Properties, ISOLATION Levels, Locks.

Transact-SQL (T-SQL)

Module 15: Basics of T-SQL Programming

- Variables Introduction & Types of Variables and Usage
- IF – Else, While structure programs.
- System functions

Module 16: Working with Stored Procedures (SP/Procs)

- Introduction to stored procedures
- Syntax of Creating, Executing, Altering, Dropping of Procedures
- Input–Output and Optional Parameters
- Procedure Optimization techniques
- Error Handling in Stored Procedures
- Stored procedures **vs.** User defined functions

Module 17: Working with Functions (UDF)

- User Defined Functions and types Of UDF
- Creating, Altering and Dropping Functions
- Scalar Functions, Inline Table Valued Functions, Multi Statement Table Valued Functions

Module 18: Implementing Triggers

- Introduction to triggers
- Constraints vs Triggers
- Creating, Altering, Dropping triggers
- Magic Tables
- Instead of Triggers

Module 19: Working with Cursors

- Introduction of cursors , Advantages of cursors, Creating Cursors, Types of cursors, Locks on cursors

Module 20: Backup and Restore

- Generating & executing SQL Script
- Taking database Backup and Restoring database using backup

Module 21: Query Performance Techniques & XML Data Types

- Query Optimization Techniques and SQL Profiler Overview.

Module 22: SQL Server Agent (Job Schedules)

- Introduction of SQL Server Agent
- Create a Job and Schedule a time to Process data automatically

Module 23: REAL-TIME PROJECT on BANKING / Retail Domain

THANK YOU ☺